Seasonal Inventory Management

1. INTRODUCTION

ToyJoy is a toy retail company operating through both physical stores and an online channel. Over the last holiday season, management observed two major operational challenges: **excess inventory for certain products** and **frequent stockouts for others**. Excess inventory leads to blocked working capital, higher storage costs, and potential product obsolescence, while stockouts result in lost sales and reduced customer satisfaction.

The purpose of this analysis is to use **historical sales and inventory data from the last three years (2023–2025)** to evaluate ToyJoy's inventory performance. By identifying patterns in product demand, seasonality, and inventory turnover, the analysis aims to support **data-driven inventory planning decisions**.

This report presents an end-to-end analytical approach covering **Exploratory Data Analysis (EDA), KPI evaluation, dashboard insights, and actionable recommendations** to optimize inventory levels, minimize waste, and improve cash flow.

Table view

Date	Year	Month	Season	Product_ID	Product_Name	Category	Store_ID	Units_Sold	Avg_Inventory	Units_In_Stock
Sunday, January 1, 2023	2023	Jan	Normal	P001	Action Figure	Action Toys	S001	320	900	580
Wednesday, February 1, 2023	2023	Feb	Normal	P001	Action Figure	Action Toys	S001	300	880	580
Wednesday, March 1, 2023	2023	Mar	Normal	P001	Action Figure	Action Toys	S001	350	850	500
Sunday, October 1, 2023	2023	Oct	Festival	P001	Action Figure	Action Toys	S001	800	1000	200
Friday, December 1, 2023	2023	Dec	Festival	P001	Action Figure	Action Toys	S001	1200	1100	-100
Monday, January 1, 2024	2024	Jan	Normal	P002	Building Blocks	Educational	S002	280	1000	720
Saturday, June 1, 2024	2024	Jun	School Holiday	P002	Building Blocks	Educational	S002	700	950	250
Sunday, December 1, 2024	2024	Dec	Festival	P002	Building Blocks	Educational	S002	1300	1200	-100
Wednesday, February 1, 2023	2023	Feb	Normal	P003	Remote Car	Electronic	ONLINE	150	800	650
Saturday, July 1, 2023	2023	Jul	Normal	P003	Remote Car	Electronic	ONLINE	500	850	350
Wednesday, November 1, 2023	2023	Nov	Festival	P003	Remote Car	Electronic	ONLINE	900	1000	100
Friday, March 1, 2024	2024	Mar	Normal	P004	Dolls Set	Soft Toys	S001	220	700	480
Thursday, August 1, 2024	2024	Aug	Normal	P004	Dolls Set	Soft Toys	S001	600	720	120
Sunday, December 1, 2024	2024	Dec	Festival	P004	Dolls Set	Soft Toys	S001	1100	900	-200
Wednesday, January 1, 2025	2025	Jan	Normal	P005	Puzzle Games	Educational	S002	200	650	450
Sunday, June 1, 2025	2025	Jun	School Holiday	P005	Puzzle Games	Educational	S002	550	700	150
Monday, December 1, 2025	2025	Dec	Festival	P005	Puzzle Games	Educational	S002	1000	850	-150

Inventory_Turnover	Stockout_Days	Unit_Cost	Unit_Price	Revenue	Excess_Inventory_Units	Excess_Inventory_Value
0.36	2	200	350	112000	580	116000
0.34	3	200	350	105000	580	116000
0.41	1	200	350	122500	500	100000
0.8	8	200	350	280000	200	40000
1.09	12	200	350	420000	0	0
0.28	1	300	550	154000	720	216000
0.74	6	300	550	385000	250	75000
1.08	10	300	550	715000	0	0
0.19	0	1500	2500	375000	650	975000
0.59	4	1500	2500	1250000	350	525000
0.9	9	1500	2500	2250000	100	150000
0.31	0	250	450	99000	480	120000
0.83	5	250	450	270000	120	30000
1.22	11	250	450	495000	0	0
0.31	1	180	300	60000	450	81000
0.79	6	180	300	165000	150	27000
1.18	10	180	300	300000	0	0

2. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted to understand sales behavior, inventory levels, and revenue contribution across products, categories, stores, and seasons.

2.1 Dataset Overview

- Time period: **Last 3 years (2023–2025)**
- Granularity: **Monthly**
- Key dimensions:
  - Product, Category, StoreID
  - Season (Festival, School Holiday, Normal)
- Key measures:
  - Units Sold
  - Average Inventory
  - Units in Stock
  - Revenue
  - Excess Inventory Value
  - Stockout Days

Data

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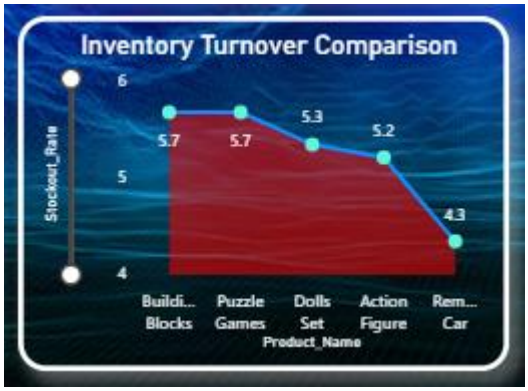
  - Units_In_Stock
  - Units_Sold
  - Year

2.2 Sales Distribution

- Sales are **not evenly distributed** across products.
- Products like **Action Figure, Remote Car, and Building Blocks** contribute a significant share of total units sold.
- Educational and Electronic categories generate **higher revenue per unit**, while Action and Soft Toys show **higher volume but lower margins**.

2.3 Inventory Behavior

- Several products maintain **high average inventory even during low-demand months**, indicating overstocking.
- Negative UnitsInStock values during peak months indicate **stockout situations**, confirming operational inefficiencies.



2.4 Seasonality Patterns

- Clear sales spikes are observed during:
    - **Festival season (Oct–Dec)**
    - **School holidays (May–Jun)**
- Normal months show relatively stable but lower demand.

EDA confirms that **seasonality plays a critical role** in ToyJoy's demand patterns and must be incorporated into inventory planning.

3. METHODS

3.1 Data Preparation

- Data was prepared using a structured CSV file and cleaned in Power BI.
- Missing attributes such as **StoreID, UnitsInStock, and Season** were derived using reasonable business assumptions.
- Blank rows were removed, and data types were standardized.

3.2 KPI Calculations

The following KPIs were computed using DAX measures in Power BI:

- **Total Units Sold**
- **Inventory Turnover Rate** = Units Sold ÷ Average Inventory
- **Stockout Rate** = Average Stockout Days
- **Excess Inventory Value**
- **Revenue Contribution per Product**

3.3 Visualization Approach

Different visualization types were chosen based on analytical purpose:

- Cards for KPIs
- Bar and column charts for comparisons
- Line charts for trends and seasonality
- Scatter plots for identifying fast- and slow-moving products
- Slicers for interactivity (Year, Season, Category, Product, Store)

4. DASHBOARD & RESULTS

An interactive **Inventory Optimization Dashboard** was developed in Power BI to address all business questions.

4.1 KPI Performance

- Total Units Sold and Revenue provide a high-level view of business performance.
- Inventory Turnover Rate highlights efficiency of stock utilization.
- Excess Inventory Value reveals capital tied up in unsold goods.

4.2 Overstock Analysis

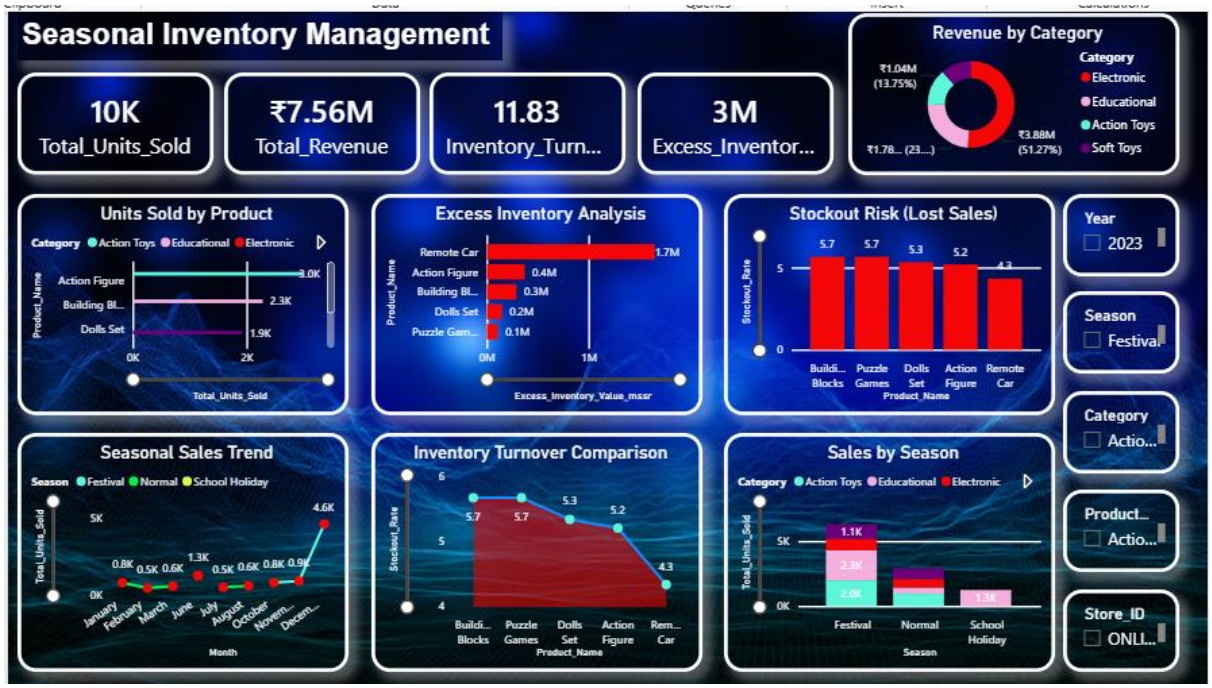
- Certain products consistently show **high excess inventory value**, indicating over-purchasing or weak demand.
- These products are prime candidates for inventory reduction or promotional clearance.

4.3 Stockout Analysis

- High stockout rates are observed for fast-selling products during festival and holiday seasons.
- This confirms that ToyJoy is **under-estimating peak demand**, leading to lost sales.

4.4 Seasonal Trends

- Line and stacked charts clearly show demand surges during festivals and school holidays.
- Products respond differently to seasons, indicating the need for **product-specific seasonal planning**.



5. INSIGHTS

Key insights derived from the analysis include:

1. **Mismatch Between Demand and Inventory**

    - Some low-selling products are overstocked, while high-demand products experience stockouts.

2. **Strong Seasonal Demand**
    - Festival and school holiday periods drive a large portion of annual sales.
    - Ignoring seasonality leads to both overstock and understock situations.
3. **Inventory Turnover Variability**
    - High turnover products indicate strong demand but insufficient replenishment.
    - Low turnover products signal slow movement and excess inventory risk.
4. **Revenue Concentration**
    - A small set of products contributes disproportionately to total revenue, emphasizing the need for focused inventory prioritization.

6. RECOMMENDATIONS

6.1 Quick Wins (Short-Term Actions)

- Reduce inventory levels for consistently low-turnover products.
- Reallocate warehouse space to high-demand, high-revenue products.
- Increase stock levels before known peak seasons to reduce stockouts.
- Introduce targeted discounts for excess inventory items.

6.2 Long-Term Strategies

- Implement **seasonal demand forecasting models** using historical trends.
- Define **dynamic reorder points** based on season, product category, and sales velocity.
- Continuously monitor inventory turnover and stockout rates using dashboards.
- Align procurement planning with promotional and festival calendars.

These actions will help minimize waste, prevent lost sales, and improve overall cash flow.

7. CONCLUSIONS

This analysis demonstrates that ToyJoy's inventory challenges are primarily driven by **misaligned stocking decisions and insufficient consideration of seasonality**. By leveraging historical data and interactive dashboards, ToyJoy can clearly identify overstocked and understocked products, understand demand patterns, and make informed inventory decisions.

The Power BI dashboard serves as a powerful decision-support tool, enabling management to continuously monitor performance and adjust strategies proactively. Implementing the recommended actions will lead to **improved inventory efficiency, reduced operational costs, and enhanced customer satisfaction**.

8. REFERENCES

1. Power BI Documentation – Microsoft
2. Inventory Management Best Practices – Industry Reports
3. Time Series & Seasonal Demand Forecasting Concepts
4. Course Material and Case Study Guidelines